

C++ > C++ 12 - strings.cpp > ...

You, 3 days ago | 1 author (You)

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1  ///! strings :
2  /*
3      ~ a string :
4          - variable contains a collection of characters
5              - surrounded by double quotes " "
6          - strings is Array of characters
7          - string is a object of the string class in C++
8          - string class is a part of the C++ Standard Library
9
10     ~ include the string library :
11         - to use the string class, we need to include the string library
12         #include <string>
13
14     ~ array :
15         - collection of items of the same type
16         - fixed size
17         - stored in contiguous memory locations
18         # to get the specific item in the array, we can use the index number of the item
19             - index starts from 0
20             variable_name[index_number]
21             - example :
22                 string my_string = "Hello, World!";
23                 char first_character = my_string[0]; // first_character will be 'H'
24                 char second_character = my_string[1]; // second_character will be 'e'
25                 char third_character = my_string[2]; // third_character will be 'l'
26                 char fourth_character = my_string[3]; // fourth_character will be 'l'
27                 char fifth_character = my_string[4]; // fifth_character will be 'o'
28                 char sixth_character = my_string[5]; // sixth_character will be ','
29
30     ~ methods :
31         * length() :
32             returns the length of the string
33             string my_string = "Hello, World!";
34             int length = my_string.length(); // length will be 13
35         * getline
36             to read a string with spaces
37             syntax :
38                 getline(cin, variable_name);
39             example :
40                 string my_string;
41                 cout << "Enter a string: ";
42                 getline(cin, my_string);
43                 cout << "You entered: " << my_string << endl;
44                 % if you have some problems with getline() function, you can use the
45                 % cin.ignore() function to ignore the newline character left in the input
46                 % buffer by the previous input operation
47                 cin.ignore(1, '\n');
48                 getline(cin, variable_name);
49
```

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50 ~ concatenation :
51     - joining two strings together
52     - we can use the + operator to concatenate two strings
53     string s1 = "Hello, ";
54     string s2 = "World!";
55     string s3 = s1 + s2; // s3 will be "Hello, World!"
56
57 ~ note:
58     ^ when use stoi function to convert string to integer,
59     ^ the string should contain only digits, otherwise it will throw an exception.
60
61 ! read string with spaces :
62     - to read a string with spaces, we can use the getline() function
63     - syntax :
64         getline(cin, variable_name);
65     - example :
66         string my_string;
67         cout << "Enter a string: ";
68         getline(cin, my_string);
69         cout << "You entered: " << my_string << endl;
70
71 */
72
73

```

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74 //~ example : 🐼
75 #include <iostream>
76 #include <string>
77 using namespace std;
78
79 int main()
80 {
81     string my_string = "ABCDEFGHIJKLMNOPQRSTUVWXYZ";
82     cout << "The length of the string is: " << my_string.length() << endl;
83
84     cout << my_string[0] << endl; // A
85     cout << my_string[6] << endl; // G
86     cout << my_string[12] << endl; // M
87
88
89     string s1="10" , s2="20";
90     string s3 = s1 + s2; // concatenation
91     cout << s3 << endl; // 1020
92
93
94     int sum = stoi(s1) + stoi(s2); // convert string to integer and then add
95     cout << "The sum is: " << sum << endl; // The sum is: 30
96
97     string fullname;
98     cout << "Enter your full name: ";
99     getline(cin, fullname); // read string with spaces
100    cout << "You entered: " << fullname << endl;
101    return 0;
102 }

```